

Uncertainty-Aware Dynamic Learning With Fuzzy-Guided Temporal Aggregation for Reliable Anomaly Detection

Dongqing Jia , Jin Yang , *Member, IEEE*, Ye Wang, Yanzhen Zhu , Huijia Liang , and Mingming Zhan

I. INTRODUCTION

Abstract—Modern cyberattacks employ stealthy, long-lived techniques that often evade traditional detectors. System provenance graphs offer rich context for anomaly detection, but deterministic node representations often struggle with uncertainty, obscuring semantics and reducing interpretability. We propose an uncertainty-aware dynamic graph learning framework with fuzzy-guided aggregation for reliable system anomaly detection. Audit logs are modeled as heterogeneous temporal graphs to preserve fine-grained features. Rather than a naive fusion of fuzzy logic and graph models, a task-specific, uncertainty-aware training principle is adopted that injects a statistically grounded fuzzy-confidence prior into both message passing and temporal aggregation. First, a Fuzzy Confidence Evaluation Matrix (FCEM) quantifies multi-scale rarity and local anomaly evidence. Second, a Dynamic Fuzzy Mechanism Aggregation (DFMA) implements a differentiable, rule-inspired temporal fusion with sparse rule activation for history reconstruction. Theoretical analysis proves DFMA's boundedness and gradient stability under mild assumptions, and explains how the fused design mitigates spurious score accumulation during long runs. Empirical evaluation on representative datasets demonstrates that this method significantly improves the detection of stealthy and long lived anomalies. It reduces the accumulation of false positives and maintains stable thresholds during extended operation while achieving competitive scores on threshold independent metric. The proposed approach balances interpretability, theoretical robustness, and operational reliability, demonstrating suitability for reliable anomaly detection in real world.

Index Terms—Reliable anomaly detection, advanced persistent threats, self-supervised graph representation learning, provenance graphs.

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THE digitalization of information has expanded cyberspace while heightening exposure to security threats. Stealthy and persistent tactics hinder timely detection. System-level provenance graphs offer a core substrate for modeling host behavior and separating benign from malicious or faulty activities [1], [2], [3], [4]. Yet ground-truth sparsity and heterogeneous interactions introduce substantial uncertainty: deterministic node/edge representations blur context, impede interpretation, and destabilize operation under workload drift.

Graph deep learning underpins many provenance-based IDSs, from graph/node classification and subgraph matching [5] to temporal encoders that capture subtle behavioral changes [3], [6]. Snapshot sequences ease temporal modeling [7] but discretization breaks continuity and masks fine-grained dynamics [8]. While temporal graphs help [3], [9], yet deployments face two gaps: (i) uncertainty in attributes, edge semantics, and evolving topology, amplified by outages/log interruptions [10] and open-world mimicry [11]; and (ii) long-run reliability, where miscalibration accumulates false positives and destabilizes thresholds.

To address these issues, we propose *FuzzyWatcher*, an uncertainty-aware dynamic graph learning framework for reliable anomaly detection on audit-log-derived provenance graphs. Audit logs are modeled as heterogeneous temporal graphs; training is benign-only via self-supervised link prediction to learn latent behavior patterns without attack labels. The key idea is to adopt a neuro-fuzzy stance: (i) bounded, monotone membership functions map multi-scale rarity and local outlier statistics to $[0,1]$ confidence degrees; (ii) these degrees act as a statistically grounded prior that modulate message passing and temporal fusion; and (iii) decisions are auditable via per-statistic membership curves, per-edge/node confidence weights, and per-window activation traces.

The main contributions of this work are summarized as follows:

- We propose *FuzzyWatcher*, an uncertainty-aware intrusion detection framework based on provenance that focuses on epistemic uncertainty in the learned benign behaviour model. It adopts a neuro-fuzzy design that injects a statistically grounded fuzzy-confidence prior into message passing and temporal aggregation, enabling robust handling of uncertainty in open, heterogeneous, and attack-label-free

environments without relying on explicit rule-based fuzzy inference.

- We design two complementary modules to ensure reliable and interpretable learning. The Fuzzy Confidence Evaluation Matrix (FCEM) assigns data driven confidence weights based on multi scale rarity and local outlier statistics to modulate node and message influence. The Dynamic Fuzzy Mechanism Aggregation (DFMA) performs differentiable and rule inspired temporal fusion with sparse activations. It reconstructs historical context while ensuring boundedness and gradient stability.
- We propose a principled anomaly scoring strategy that operationalizes anomalousness, rarity, and propagation at the window level, coupled with extreme-value-based thresholding for stable calibration under drift. On representative provenance benchmarks and real audit logs, *FuzzyWatcher* enhances detection of long-lived stealthy anomalies, reduces long-run false-positive accumulation, demonstrating both effectiveness and deployability.

The rest of the paper is structured as follows. Section II reviews related work. Section IV details our framework. Section V and Section VI present experimental results and comparative analysis. Section VII concludes.

II. RELATED WORK

This paper focuses on provenance graph-based anomaly detection for host systems. The key is to build a dynamic behavior relationship graph by tracking and analyzing the interaction history of entities in the system, thereby identifying malicious behaviors that deviate from normal patterns. In what follows, we review provenance-based intrusion/anomaly detection (PID), anomaly detection, and uncertainty analysis.

PIDSs are typically divided into rule-based and anomaly-based approaches [12]. Rule-based methods match predefined rules to detect known patterns [13], [14], often relying on supervised labels and prior knowledge for graph construction [1]; they work on known threats but generalize poorly, and labeling plus delayed rule updates hinder scalability [15]. Anomaly-based methods train on benign activity and flag deviations [10], [16]. Free from prior attack knowledge, they are more adaptable to unseen threats. For instance, Threatscore statistical irregularities in entities and interactions [10]. Yet shallow representations can miss rare or evolving benign behaviors, raising false positives [2]; attackers may mimic frequent benign patterns (e.g., www.yale.edu) to evade detection without richer context [17].

Given stealthy multi-stage attacks, anomaly-based strategies dominate. Recent learning-based systems employ snapshot sequences to preserve order, though risking continuity breaks [7], [8]. Temporal dataflow graphs can retain event semantics. Representative works include recommendation-style detection (ShadeWatcher [18]), rarity-aware scoring (KAIRIS [3]), self-supervised reconstruction (MAGIC [8]), long-horizon reasoning (TaGAPT [19]), and hybrid rule distillation for accuracy-latency-cost trade-offs [6], [20], but most assume deterministic representations and offer limited treatment of uncertainty. In parallel, general graph anomaly detection has advanced via dynamic-graph models for distribution shift [21] and contrastive

TABLE I
DETECTION PERFORMANCE OF *FuzzyWatcher* ON ALL DATASETS.
E5-CADETS+: DETECTS FALSE POSITIVES CAUSED BY POST-ATTACK
RESIDUALS; E5-CADETS-: TREATS RESIDUALS AS ATTACK INSTANCES

Datasets	TP	TN	FP	FN	Recall	F1	AUC
E3-THEIA	8	126	1	1	0.889	0.889	0.985
E3-CADETS	4	174	1	0	1.000	0.889	0.997
E3-ClearScope	5	112	2	0	1.000	0.833	0.991
E5-THEIA	2	82	0	0	1.000	1.000	1.000
E5-CADETS+	5	247	0	2	0.714	0.833	0.857
E5-CADETS-	15	238	0	1	0.938	0.968	0.969
E5-ClearScope	10	289	4	0	1.000	0.833	0.993
StreamSpot	100	375	0	0	1.000	1.000	1.000

methods for structural imbalance. Related non-graph detectors tackle long-range dependence and imbalance using multiscale attention, cluster resampling, adaptive aggregation, causal regularization, and recurrent VAEs [22], [23], [24], [25], [26].

Uncertainty quantification in graph neural networks improves reliability through epistemic/aleatoric estimation using ensembles, Bayesian methods, or MC-dropout, alongside post-hoc calibration [27]. These advancements highlight gaps in current security pipelines and motivate our development of provenance-based detection that is both dynamic-aware and explicitly uncertainty-conscious.

Evolving Fuzzy Systems (EFSs) are online self-adaptive models that update structure and parameters to track nonstationary streams while providing interpretable IF- THEN rules [28], [29]. These neuro-fuzzy architectures enable transparent reasoning under uncertainty for real-time analytics. We adopt a data-driven design where similarity-based antecedents generate statistics-grounded confidence weights for message passing. Additionally sparsity-inducing rule activations support temporal aggregation to ensure both efficiency and reliability (see Section IV-B).

Building on this perspective, we propose a window-level detector for heterogeneous environments focusing on uncertainty calibration and long-run stability. Trained exclusively on benign data, the model injects fuzzy-confidence priors into message passing and aggregation to mitigate noisy dependencies. By applying an extreme-value-based threshold to stabilize the operating point, our approach enables earlier detection of sustained anomalies while remaining robust to unseen behaviors.

III. SYSTEM AND THREAT MODEL

A. Threat Model

Similarly to recent advances in PIDs [3], [6], [8], this work considers both external intrusions and persistent malicious activities within the system. The adversary is assumed to target sensitive internal information using complex and novel intrusion techniques. Rather than launching overt attacks, malicious actions are deliberately concealed within large volumes of benign activity. As *FuzzyWatcher* follows an anomaly-based detection paradigm, the training provenance graphs are assumed to represent trustworthy, attack-free system behavior. The framework relies on learning normal patterns from benign data and does not require knowledge of specific attack signatures. The threat

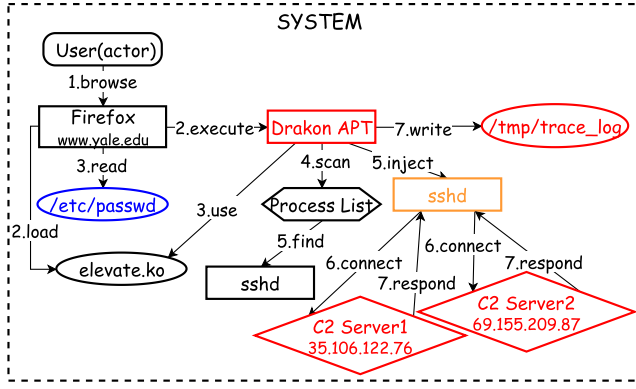


Fig. 1. Motivating provenance subgraph from DARPA TC-E5 (THEIA). Rectangles, ellipses, diamonds, and hexagons represent processes, files, sockets, and system calls, respectively. Red = malicious entities/tainted artifacts and C2 endpoints; orange = compromised but originally benign components blue = sensitive assets; black = benign context. Unrelated benign entities and interactions are pruned for clarity.

model excludes hardware-level, kernel-level, and side-channel attacks. The underlying hardware, operating system, and auditing infrastructure are assumed to be part of the trusted computing base (TCB) and free from compromise.

B. Provenance Graph and Problem Setting

We model audit log-derived behaviors as a provenance graph. It is a heterogeneous temporal graph $G = (V, E, T)$ with typed entities $\mathcal{T}_v \in \{\text{process}, \text{file}, \text{socket}\}$ and interactions $\mathcal{T}_e \subseteq \{\text{exec}, \text{read}, \text{write}, \text{connect}, \dots\}$. Each event is $e_i = (u_i, r_i, v_i, t_i)$ where $u_i, v_i \in V$, $r_i \in \mathcal{T}_e$, $t_i \in \mathbb{R}_+$. We partition time into fixed windows indexed by n , denoted as $W_n = [\tau_n, \tau_n + \delta)$. The whole framework adopts a benign-only training regime with a self-supervised link-prediction to learn normal dynamics. Let $s(e_i)$ denote per-event scores; a window score $S(W_k)$ is obtained using a monotone fusion operator $\mathcal{A}$ over $\{s(e_i) : e_i \in W_n\}$, i.e. $S(W_n) = \mathcal{A}(\{s(e_i)\})$. The higher $S(W_n)$ indicates a greater departure from normal behavior. Operational decisions are ultimately driven by our anomaly scoring strategy, whose core objective is to surface sustained departures from normal behavior as early as possible in open and heterogeneous environments.

Running Example: Fig. 1 shows a representative provenance subgraph of DARPA TC-E5 (THEIA). A user's browsing session to a compromised benign site is followed by injection of driver-assisted code into trusted processes (e.g., sshd), establishing covert C2 while interleaving with routine operations. Because the activity closely mimics normal workflows and unfolds over an extended period, static signatures and snapshot-style detectors tend to miss it: each short slice appears benign and locally rare benign bursts further confound decision rules. This example motivates the requirement for reasoning about the long-term temporal context and uncertainty.

IV. SYSTEM DESCRIPTION

Although adversaries can mimic benign routines and hide within long stretches of normal activity, sustained campaigns

inevitably leave structural (who interacts with whom) and temporal (when/how often) fingerprints that deviate from habitual behavior. *FuzzyWatcher* casts these deviations into a neuro-fuzzy rule view on dynamic provenance graphs: prototype-driven antecedents provide auditable soft memberships that weight evidence, trainable consequents encode local and historical context, and normalized activations aggregate multi-scale signals. Concretely, we operationalize this by (i) injecting a statistics-grounded fuzzy-confidence prior (FCEM) that discounts ubiquitous carriers while elevating locally rare but consistent evidence, and (ii) performing bounded, rule-inspired temporal fusion (DFMA) with sparsity-inducing activations to preserve long-run stability under drift.

In this work, we focus exclusively on epistemic uncertainty. *FuzzyWatcher* quantifies how well a node or window is supported by past benign observations in the learned provenance-graph representation. Nodes that lie in low-density regions of this benign manifold, or that trigger unstable rule activations, are treated as epistemically uncertain, reflecting behaviours for which the model has little prior knowledge. We do not explicitly model aleatoric uncertainty (e.g., noise in audit logs, missing events).

Fig. 2 depicts the end-to-end pipeline: audit logs form heterogeneous temporal provenance graphs (Section III); a temporal context encoder and a global topology encoder produce multi-scale event representations; FCEM supplies antecedent-side memberships to modulate message passing (Section IV-B1); DFMA instantiates temporal rule aggregation over learnable consequents (Section IV-B2); window-level anomaly scoring and threshold calibration then stabilize operations (Section IV-C).

A. Temporal and Topological Backbone

1) *Dynamic Contextual Modeling:* Temporal dependency is essential for provenance-based anomaly detection: static graphs miss burstiness, order, and drift. We therefore adopt a Temporal Graph Network (TGN) [30] to model event-driven interactions with node-specific memory. For node v , the memory is updated at event time t as

$$\mathbf{M}_v(t) = \text{GRU}(\mathbf{M}_v(t^-), \bar{\mathbf{m}}_v(t)), \quad (1)$$

where $\bar{\mathbf{m}}_v(t)$ aggregates messages from recent interactions. Let $\mathcal{N}(v, t)$ be the set of events involving v before t ; a time-decay kernel $\phi(\Delta t) = \exp(-\beta \Delta t)$ emphasizes recency, and the aggregated message is

$$\bar{\mathbf{m}}_v(t) = \sum_{(u, r, t') \in \mathcal{N}(v, t)} \phi(t - t') \cdot \text{MSG}(\mathbf{x}_u, \mathbf{x}_v, \mathbf{x}_r, \Delta t). \quad (2)$$

This captures local temporal causality so that benign bursts (e.g., backups) and stealthy deviations (e.g., lateral movement) are not smoothed into indistinguishable patterns. In practice, identifiers may stay constant while behaviors drift; tracking temporal deviations is therefore more informative than static features alone.

2) *Topological Augmentation:* While TGN captures local temporal patterns, global structural context is needed for multi-hop behaviors. *FuzzyWatcher* integrates an inductive

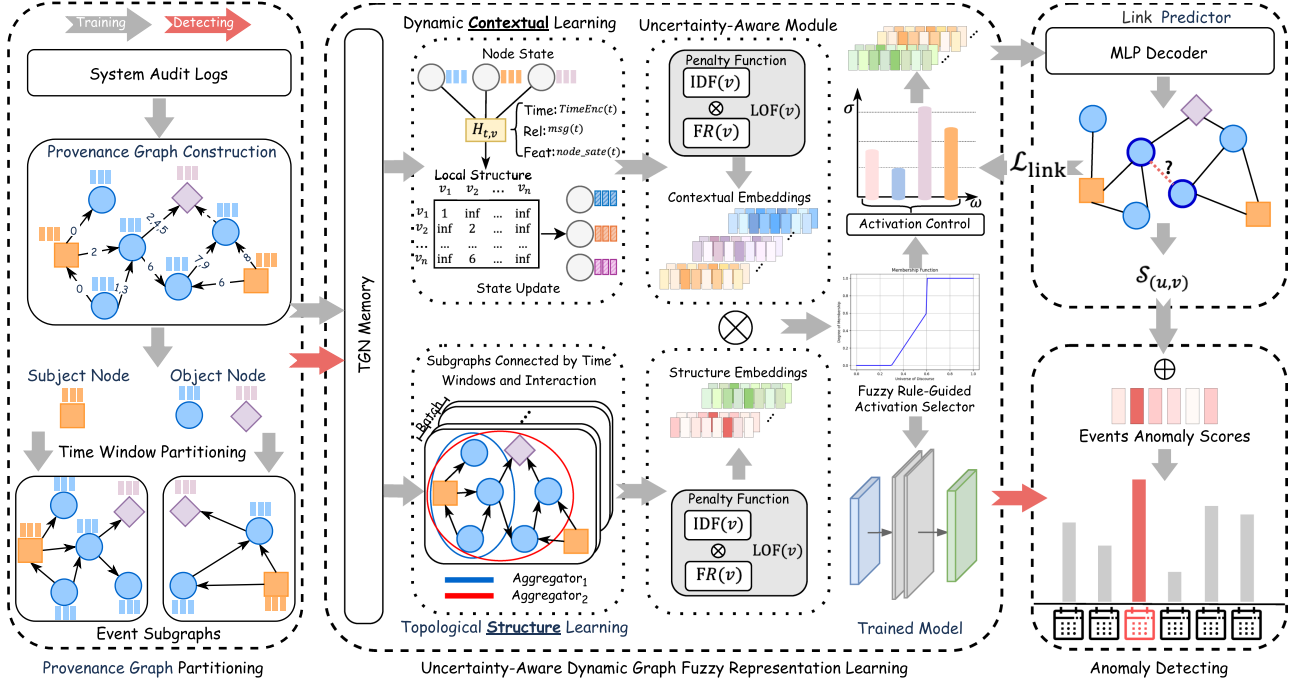


Fig. 2. Overall architecture of the *FuzzyWatcher* framework, illustrating the major modules and the self-supervised anomaly detection workflow.

neighborhood-aggregation encoder (GraphSAGE) [31]. At layer l , node i is updated by mean aggregation:

$$\mathbf{H}_i^{(l)} = \sigma \left(\mathbf{W}^{(l)} \left[\mathbf{H}_i^{(l-1)} \parallel \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{H}_j^{(l-1)} \right] \right). \quad (3)$$

This augmentation provides long-range, structurally consistent cues while remaining scalable and inductive on evolving provenance graphs. Coupled with temporal modeling, the backbone captures hybrid local-global structures at appropriate receptive-field depths.

B. Uncertainty-Aware Learning Module

To detect rare and long-lived threats under nonstationarity, we represent and propagate uncertainty through a neuro-fuzzy rule: FCME estimates antecedent memberships (a confidence measure of the uncertainty in the current evidence), and DFMA performs temporal rule activation and aggregation over trainable consequents, thereby modulating message passing and fusion. Together, they expose auditable signals (memberships and activations) while remaining fully differentiable and data-driven.

1) *Fuzzy Confidence Evaluation Matrix (FCME)*: *Fuzzy-Watcher* is only trained on benign data, where rare but legitimate behaviors can inflate false positives. Unlike fixed node importance, FCME is learned and adaptive: it updates online from benign-only evidence and reflects how often, how regularly, and where a node appears, thereby capturing different sources of epistemic uncertainty in node behavior, yielding memberships that temper noisy carriers, yet preserve legitimate rarity. FCME avoids overfitting to superficial rarity by requiring

structural consistency as a prerequisite for assigning high importance. This allows the model to distinguish between under-observed but stable patterns and structurally inconsistent behaviors.

FCME is guided by three factors: temporal rarity, feature irregularity, and structural deviation.

$$\text{IDF}(v) = \log \frac{N}{1 + |\{n : v \in \mathcal{N}_n\}|}, \quad (4)$$

$$\text{FR}(v) = \frac{1}{d} \sum_{i=1}^d \mathbb{I}(x_{v,i} \notin \mathcal{C}_i), \quad (5)$$

$$\text{LOF}(v) = \frac{1}{|\mathcal{N}(v)|} \sum_{j \in \mathcal{N}(v)} \frac{k - \text{dist}(j)}{k - \text{dist}(v)}, \quad (6)$$

where N is the total number of windows, $\mathcal{N}_n$ denotes the active-node set in the n -th window, where n indexes the windows. In FCME, windows are activity-based, each containing a comparable number of active nodes or events. $\mathcal{C}_i$ is the benign coverage range of feature i (estimated on training data), and $k - \text{dist}(\cdot)$ follows the standard definition of LOF [32], [33]. All statistics are defined with respect to the benign reference learned during training. Specifically, IDF reflects temporal rarity relative to high-frequency activity, FR measures feature deviation from benign coverage ranges, and LOF quantifies structural deviation from local benign interaction patterns. The statistics are fused into a scalar fuzzy confidence:

$$\text{Conf}_v^{(n)} = \sigma(w_1 \text{IDF}(v) + w_2 \text{FR}(v) - w_3 \text{LOF}(v)), \quad (7)$$

with learnable weights w_1, w_2, w_3 and sigmoid $\sigma(\cdot)$.

Epistemic uncertainty at node-level is modeled by a unified statistical profile defined by temporal rarity (IDF), feature irregularity (FR), and structural deviation (LOF). Let $\Phi(v) =$

$[\text{IDF}(v), \text{FR}(v), -\text{LOF}(v)]^\top$ denote the statistical profile. To ensure causal consistency in the streaming setting, $\text{Conf}_v^{(n)}$ is derived from historical observations prior to n -th window, and then assigned to all events with timestamps $t \in n$. $\text{Conf}_v^{(n)}$ is not a measure of reliability in the conventional sense, but rather serves as a proxy for epistemic uncertainty. Higher values indicate under-observed but potentially informative patterns, while lower values correspond to well-supported and frequently observed patterns. FCEM can thus be interpreted as a data-driven membership function that maps statistical features to soft memberships for uncertainty-aware representation learning.

Benign activities tend to follow stable operational routines and exhibit consistent interaction patterns, whereas malicious activities are goal-driven and more likely to introduce structural inconsistencies even when mimicking benign carriers. Accordingly, FCEM suppresses nodes that exhibit local structural inconsistency through the LOF penalty, even when they appear infrequently, preventing overreaction to transient noisy observations.

During training, these statistics modulate node contributions to message passing and learning updates. Nodes with high IDF and FR but low LOF correspond to rare yet structurally consistent behaviors and are emphasized, whereas nodes with elevated LOF are suppressed to avoid unreliable updates. This encourages the model to focus on informative but under-observed patterns while maintaining robustness to noise and structural anomalies.

All intermediate signals IDF/FR/LOF and $\text{Conf}_v^{(n)}$ are log-gable. We directly use $\text{Conf}_v^{(n)}$ as the antecedent membership to steer subsequent message passing and temporal fusion. (12), (19).

FCEM is lightweight. IDF and FR update per window via counters and feature-range tracking, while LOF is computed on local neighborhoods. The online cost is $O(|\mathcal{N}(v)|)$ per active node and adds negligible latency to the backbone.

During inference, the same statistics are computed online without requiring labels, and mapped through the learned function $\Phi(v) \mapsto \text{Conf}_v^{(n)}$. This yields an uncertainty-aware weighting signal that is injected into temporal fusion and intermediate anomaly estimation, tempering transient or structurally inconsistent signals while preserving sustained and coherent deviations. FCEM thus maintains consistent uncertainty-aware behavior across training and deployment without additional supervision or inference overhead.

2) *Dynamic Fuzzy Mechanism Aggregation (DFMA)*: To reconstruct temporal context under drift, DFMA softly activates historical states via rule-inspired weights. DFMA operates at the event level, while incorporating window-level uncertainty through $\text{Conf}_v^{(n)}$. At each time t , DFMA treats the K most recent embeddings as trainable consequents and produces a fuzzy-weighted summary, where historical states are selectively activated based on their agreement with the current embedding and neighborhood context.

For node v , maintain a temporal embedding sequence

$$\mathcal{H}_v = \left\{ \mathbf{h}_v^{(t,K)}, \dots, \mathbf{h}_v^{(t,1)} \right\}. \quad (8)$$

The temporal aggregation output of DFMA is defined by

$$\mathbf{z}_v^t = \sum_{k=1}^K \gamma_k(v, t) \mathbf{h}_v^{(t,k)}, \quad (9)$$

with normalized firing strengths

$$\gamma_k(v, t) = \frac{\exp(-\lambda_k D_k)}{\sum_{k'=1}^K \exp(-\lambda_{k'} D_{k'})}. \quad (10)$$

The composite distance balances representation drift and neighborhood divergence:

$$D_k = \alpha \left\| \mathbf{h}_v^t - \mathbf{h}_v^{(t,k)} \right\|^2 + \beta \text{JS}(\mathcal{N}_t(v), \mathcal{N}_{(t,k)}(v)), \quad (11)$$

with $\alpha, \beta \geq 0$ and $\text{JS}(\cdot, \cdot) \in [0, \log 2]$. We adopt JS divergence to compare neighborhood distributions, as $\mathcal{N}_t(v)$ is represented as a discrete and heterogeneous distribution over interaction types. JS operates directly on probability mass functions and captures differences in structural composition, making it more suitable than vector-based similarities for this setting. The weights $\{\lambda_k\}$ control the sharpness of the softmax and thus the effective sparsity of activations. Since the distance is used within a softmax, the aggregation primarily depends on the relative ordering of distances rather than their absolute values, which makes the mechanism robust to the exact choice of divergence measure. Finally, an uncertainty-guided interpolation yields the updated representation

$$\tilde{\mathbf{h}}_v^t = \text{Conf}_v^{(n)} \cdot \mathbf{h}_v^t + (1 - \text{Conf}_v^{(n)}) \cdot \mathbf{z}_v^t. \quad (12)$$

DFMA assigns similarity-aware activations $\{\gamma_k\}$ that respond to both embedding drift and neighborhood change: small D_k yields higher activation, large D_k suppresses it. This ensures that only temporally and structurally consistent historical states contribute to the aggregation. The resulting summary $\mathbf{z}_v^t$ preserves consistent temporal evidence while suppressing transient noise. Sharpness parameters $\{\lambda_k\}$ encourage sparse selections, mitigating overfitting to history without extra message-passing depth. The distances $\{D_k\}$ and activations $\{\gamma_k\}$ are auditable, and computing $\mathbf{z}_v^t$ costs $O(Kd)$ per node per step, where d is the embedding dimension.

From a fuzzy-rule perspective, recent self-states $\{\mathbf{h}_v^{(t,k)}\}_{k=1}^K$ act as temporal prototypes (trainable consequents) for node v . We first collect the information available at time t into a joint descriptor

$$\mathbf{x}_v^t \triangleq (\mathbf{h}_v^t, \mathcal{N}_t(v), \Phi(v)), \quad (13)$$

which summarizes the current embedding, neighborhood and statistical profile of v . For each lag k , DFMA then induces a local temporal rule of the canonical form

$$\mathcal{R}_{v,k} : \text{IF } \mathbf{x}_v^t \in \mathcal{A}_{v,k}^t \text{ THEN } \mathbf{c}_{v,k}^t = \mathbf{h}_v^{(t,k)}, \quad (14)$$

where the antecedent fuzzy set $\mathcal{A}_{v,k}^t$ encodes low-uncertainty and similar to the k -step prototype. Its membership function decomposes as

$$\mu_{v,k}^t = \mu_{\mathcal{A}_{v,k}^t}(\mathbf{x}_v^t) = (1 - \text{Conf}_v^{(n)}) \gamma_k(v, t), \quad (15)$$

This formulation couples uncertainty and similarity, so that historical states contribute only when they are both consistent with the current observation and supported under the uncertainty-aware confidence. This product form $(1 - \text{Conf}_v^{(n)}) \gamma_k(v, t)$ plays the role of a product t-norm that combines uncertainty and similarity in the antecedent membership. with similarity quantified by the composite distance $D_k(v, t)$ in (11) inside the softmax weights $\gamma_k(v, t)$. Hence $\mu_{v,k}^t$ is large only when the current state of v is both low-uncertainty and close to the k -step historical prototype in representation and neighborhood space.

Aggregating the rule consequents with normalized activations recovers the history summary,

$$\mathbf{z}_v^t = \sum_{k=1}^K \gamma_k(v, t) \mathbf{c}_{v,k}^t = \sum_{k=1}^K \gamma_k(v, t) \mathbf{h}_v^{(t,k)}, \quad (16)$$

and the uncertainty-guided interpolation in (12) implements the overall rule action: when $\text{Conf}_v^{(n)}$ is high, $\tilde{\mathbf{h}}_v^t$ primarily preserves the current embedding $\mathbf{h}_v^t$, allowing informative deviations to remain visible in the representation; when $\text{Conf}_v^{(n)}$ is low, DFMA incorporates temporally consistent historical context. Emphasizing the historical component suppresses local fluctuations and prevents minor deviations from being repeatedly written into the representation, to obtain a more stable estimate. During training, this mechanism regulates how the model balances stability and adaptation: low-uncertainty observations encourage consistency with learned temporal patterns, while high-uncertainty observations preserve informative deviations and prevent them from being prematurely smoothed.

During inference, increased reliance on historical context for low-uncertainty observations stabilizes detection and reduces false positives caused by transient perturbations, whereas preserving the current state for high-uncertainty observations helps maintain sensitivity to emerging anomalies. The sparse activation design avoids additional message-passing depth and limits the influence of redundant historical states.

Proposition 1. (boundedness and gradient stability). If $\text{Conf}_v^{(n)} \in [0, 1]$, with finite $\lambda_k \geq 0$, and $\gamma_k(v, t) \geq 0$ satisfying $\sum_k \gamma_k(v, t) = 1$, then $\tilde{\mathbf{h}}_v^t$ lies in the convex hull of $\{\mathbf{h}_v^t, \mathbf{h}_v^{(t,1)}, \dots, \mathbf{h}_v^{(t,K)}\}$ and hence $\|\tilde{\mathbf{h}}_v^t\| \leq \max_k \|\mathbf{h}_v^{(t,k)}\|$. Moreover, the Jacobian of $\tilde{\mathbf{h}}_v^t$ w.r.t. $(\mathbf{h}_v^t, \{\mathbf{h}_v^{(t,k)}\})$ is a convex combination of Lipschitz maps, yielding Lipschitz-stable gradients. Proof and extensions are in Supp.B. These properties curb spurious score accumulation over long runs, enable stable thresholds across days and domains, and mitigate drift-induced false positives, thereby showing deployability for long-running detection systems.

3) *Uncertainty-Aware Representation Learning:* We define uncertainty as epistemic ambiguity caused by evolving nodes across time in graphs: within a time interval, nodes may shift between normal and abnormal behaviors, and the same nominal entity can exhibit different patterns across windows. This

Algorithm 1: Uncertainty-Aware Training.

Require: benign temporal provenance graph $\mathcal{G}$, features x_v, x_e , history length K , parameters Θ
Ensure: trained encoder with uncertainty-aware embeddings
1: initialize TGN memories $\{M_v\}$, FCEM statistics (IDF/FR/LOF), DFMA history buffers
2: **for** the n -th training window W_n in chronological order **do**
3: **for** each event $(u, r, v, t) \in W_n$ **do**
4: **Temporal backbone:** update node memories M_u, M_v and obtain base embedding $\mathbf{h}_v^t \leftarrow \text{ENC}(M_v(t), x_v)$
5: **Update FCEM statistics:** refresh IDF/FR/LOF counters for nodes touched in W_n
6: **Fuzzy confidence (FCEM):** $s_v^t \leftarrow \text{Conf}_v^{(n)}$ // antecedent membership
7: **Temporal fusion (DFMA):** retrieve recent embeddings $\{\mathbf{h}_v^{(t,k)}\}_{k=1}^K$ and compute
8: $\mathbf{z}_v^t \leftarrow \text{DFMA_HistorySummary}(v, \{\mathbf{h}_v^{(t,k)}\})$ // rule consequents and activations
9: **Uncertainty-guided update:**
 $\tilde{\mathbf{h}}_v^t \leftarrow s_v^t \mathbf{h}_v^t + (1 - s_v^t) \mathbf{z}_v^t$
10: **Structural encoding:** pass $\tilde{\mathbf{h}}_v^t$ through GraphSAGE layers with uncertainty-aware smoothing
11: **Logging (optional):** record $\Phi(v)$, s_v^t , DFMA distances/weights for interpretability
12: **end for**
13: **Self-supervised loss:** compute link-prediction loss $\mathcal{L}$
14: update Θ by backpropagation on $\mathcal{L}$
15: **end for**

temporal variability is the target of our fuzzy prior and history fusion mechanisms.

Temporal encoding, memory updates, DFMA aggregation, and the self-supervised objective are defined over event-level interactions. Time windows indexed by n are used to compute statistical profiles in FCEM, including IDF, FR, and LOF, which provide a window-level characterization of node behavior. It's a two-level temporal structure: representation learning operates at the event level, while uncertainty estimation is performed at the window level and injected into event-level representations through $\text{Conf}_v^{(n)}$.

Recall that the statistical profile $\Phi(v)$ and uncertainty score $\text{Conf}_v^{(n)}$ defined in (7) can be compactly written as (17) shown at the bottom of this page

$$\text{Conf}_v^{(n)} = \sigma(\mathbf{w}^\top \Phi(v)), \quad (18)$$

with $\mathbf{w} = [w_1, w_2, w_3]^\top$. As in Section IV-B1, we use s_v^t and $\text{Conf}_v^{(n)}$ interchangeably to denote the antecedent membership. Let $\mathbf{h}_v^t$ be the backbone embedding at time t and $\mathbf{z}_v^t$ the DFMA

$$\mathbf{A}_v^{(l)}(t) = \mathbf{W}^{(l)} \left(\mathbf{H}_v^{(l-1)}(t) \parallel \text{AGG}_{\text{mean}} \left\{ \mathbf{H}_j^{(l-1)}(t) \mid j \in \mathcal{N}(v) \right\} \right) \quad (17)$$

history summary (9)–(10). The fused representation follows the DFMA interpolation in (12), where temporally consistent history is integrated under uncertainty-aware modulation. Specifically, low-uncertainty observations allow temporally consistent historical states to be incorporated as complementary evidence, while high-uncertainty observations preserve the current representation to avoid potential bias from inconsistent historical patterns. $\text{Conf}_v^{(n)}$ plays the role of an antecedent membership and $\mathbf{z}_v^t$ that of a consequent aggregation, (12) realizes the corresponding rule action.

For multi-layer structural encoding, we apply uncertainty-aware smoothing at every layer. Denote the pre-aggregation projection as (17), and let $\{\mathbf{H}_v^{(l)}(t, k)\}_{k=1}^K$ be the layer- l history prototypes with normalized activations $\{\gamma_k(v, t)\}$. The uncertainty-modulated layer output is (19) shown at the bottom of this page, which injects uncertainty-aware modulation into each layer and performs temporal aggregation via DFMA without increasing message-passing depth. In other words, higher antecedent membership (higher uncertainty) emphasizes the current evidence $\sigma(\mathbf{A}_v^{(l)}(t))$, while lower uncertainty allows temporally consistent historical representations to be incorporated. This design ensures that historical information is integrated conditionally, rather than being relied upon as a default reference.

Algorithm 1 summarizes the core update: the model performs temporal message passing and memory updates to obtain a base embedding, computes the statistical profile Φ and uncertainty score Conf , forms a DFMA historical summary, fuses current and historical representations, and updates parameters via a self-supervised loss.

The module is wrapped around memory-based or structural backbones with only $O(Kd)$ extra work per active node. The statistical prior and DFMA weights can be logged.

C. Anomaly Detection

Stealthy, long-lived campaigns often blend with benign activities and evade one-shot detectors. This challenge can be addressed by using a window queue that aggregates temporally adjacent windows sharing suspicious nodes [3]. *FuzzyWatcher* enhances this by assessing three campaign lifecycle dimensions. First *anomalousness* is identified when edge reconstruction error exceeds a dynamic threshold. Second *rarity* is quantified via IDF. Third *propagation* tracks the reappearance of rare suspicious nodes across windows to capture the sequential dependencies inherent in attack behaviors.

The anomaly assessment at inference time only. Detection operates over fixed-duration windows (15 minutes), distinct from the activity-based windows used in FCEM. Both follow the same temporal progression but differ in boundary definition. Let n denote the current decision window with edge set $\mathcal{E}_n$. For each edge $e \in \mathcal{E}_n$, let ℓ_e^n be the reconstruction loss. Denote the mean and standard deviation within the window as $\mu_n = \mathbb{E}[\ell^n]$ and

$\sigma_n^{(\text{std})} = \text{Std}[\ell^n]$. We quantify anomalousness by standardizing losses

$$A_e^n = \frac{\ell_e^n - \mu_n}{\sigma_n^{(\text{std})}}, \quad A_v^n = \max_{e \in \text{Inc}_n(v)} A_e^n, \quad (20)$$

and apply a robust short-term screen with a dynamic threshold

$$\tau_n = \mu_n + 1.5 \sigma_n^{(\text{std})}, \quad \mathcal{S}_n = \{v : A_v^n > \tau_n\}. \quad (21)$$

We measure rarity via a normalized inverse-document frequency,

$$\hat{R}_v = \frac{\text{IDF}(v)}{\max_{v'} \text{IDF}(v')} \in [0, 1], \quad (22)$$

and set the rarity cutoff $r_0 = \log 10 / \max_{v'} \text{IDF}(v')$:

$$\mathcal{R}_n = \{v \in \mathcal{S}_n : \hat{R}_v \geq r_0\}, \quad (23)$$

which emphasizes locally uncommon actors while excluding ubiquitous carriers.

Propagation captures temporal persistence across adjacent windows in an online manner. Let $\mathcal{S}_n = \{v : A_v^n > \tau_n\}$ be the suspicious set in window n and $\mathcal{R}_n = \{v \in \mathcal{S}_n : \hat{R}_v \geq r_0\}$ its rare-suspicious subset. A window n continues the current episode if rare suspicious nodes persist:

$$\lambda_{n+1} = \mathbb{I}[\mathcal{R}_{n+1} \cap \mathcal{R}_n \neq \emptyset] = 1. \quad (24)$$

Otherwise, a new queue is started. Equivalently, a soft variant uses a rarity-weighted overlap

$$\text{Prop}_n = \frac{\sum_{v \in \mathcal{R}_n \cap \mathcal{R}_{n-1}} \hat{R}_v}{\sum_{v \in \mathcal{R}_n \cup \mathcal{R}_{n-1}} \hat{R}_v}, \quad (25)$$

when $\sum_{v \in \mathcal{R}_n \cup \mathcal{R}_{n-1}} \hat{R}_v = 0$, $\text{Prop}_n = 0$. This formulation is causal, relying only on the current and previous windows, and thus supports online deployment.

The final window-level anomaly score integrates the three criteria:

$$\text{Score}_n = \sum_{v \in \mathcal{S}_n} \log(1 + A_v^n) \cdot \hat{R}_v \cdot (1 + \text{Prop}_n), \quad (26)$$

where bounded scaling stabilizes contributions from extreme values. To capture sustained behaviors, we form queues q by linking consecutive windows satisfying the propagation condition. The queue-level score accumulates evidence as

$$\text{Score}_q = \sum_{n \in q} \log(1 + \text{Score}_n). \quad (27)$$

which reinforces persistent signals while suppressing isolated spikes. Operational thresholding uses Peaks-Over-Threshold(POT) with extreme-value modeling(GPD). Given observed queue scores $\{s_\nu\}$ and a high baseline u , excesses $y_\nu = s_\nu - u$ are fitted with

$$F(y) = 1 - (1 + \xi y / \sigma)^{-1/\xi}, \quad (28)$$

$$\tilde{\mathbf{H}}_v^{(l)}(t) = \sigma(\mathbf{w}^\top \Phi(v)) \cdot \sigma(\mathbf{A}_v^{(l)}(t)) + (1 - \sigma(\mathbf{w}^\top \Phi(v))) \cdot \sum_{k=1}^K \gamma_k(v, t) \cdot \mathbf{H}_v^{(l)}(t, k) \quad (19)$$

yielding the $(1 - q)$ tail threshold

$$\tau = u + \frac{\sigma}{\xi} \left((1 - q)^{-\xi} - 1 \right). \quad (29)$$

Queues with $\text{Score}(q) > \tau$ are flagged for analyst inspection.

V. EVALUATION

FuzzyWatcher is implemented in Python, with preprocessing handled by `scikit-learn` and graph operations built on `PyTorch Geometric (PyG)`. We evaluate its performance on seven public datasets containing both benign and malicious activity logs, spanning Android, Linux, and FreeBSD environments (see Section V-A).

Experiments are run on a Windows 10 server with 64 GB RAM and an Intel i9-14900 K CPU at 3.20 GHz. Unless otherwise noted, we use a node embedding size of 16, memory size of 100, edge embedding size of 200, and neighborhood sampling size of 20. We use both traditional and recent dedicated evaluation metrics for attack detection as evaluation standards. See the Appendix, available online, for details.

A. Dataset

We use datasets from the DARPA Transparent Computing (TC) program and Manzoor et al. [34], [35], [36], commonly adopted for provenance-based anomaly detection. These datasets span diverse platforms and logging environments.

The StreamSpot dataset is synthetic, comprising 600 graph snapshots from six scenarios: five benign and one attack involving a Flash exploit. Malicious edges are rare across datasets, reflecting realistic threat sparsity. Due to the absence of standardized labels in DARPA data, we follow prior annotation strategies [3], [6] for evaluation. Training remains anomaly-based. For DARPA datasets, we use a temporally ordered split with three days of benign logs for training, one for validation, and three subsequent days with attacks for testing. This setup reflects the natural evolution of system activity over time and avoids artificially constructed or randomly mixed data. For StreamSpot, one benign graph per scenario is used for training, 24 for validation, and the rest for testing.

B. Detection Performance

FuzzyWatcher is trained solely on benign data and evaluated over segmented time windows, with ground truth derived from publicly released attack reports consolidated by Cheng et al. [3]. Table I reports per-dataset Accuracy, AUC, and confusion-matrix counts.

As summarized in Table I, the method achieves near-perfect recall across datasets while maintaining competitive AUC and F1. Because detection is performed at the window-level, a calibrated anomaly threshold prunes the candidate pool aggressively, shrinking tens of thousands of windows to only a few hundred for analyst triage; for example, E5-THEIA retains merely 82 benign windows (TNs), illustrating effective narrowing of the search space. A small number of false positives arise from memory residues: once a node has been compromised, its historical state can transiently raise downstream anomaly

TABLE II
ANOMALY DETECTION PERFORMANCE IN DARPA

Dataset	Method	Acc	Prec	Rec	F1	AUC
E3-THEIA	Threatrace	0.990	0.870	0.990	0.926	-
	FLASH	0.993	0.918	0.998	0.956	0.982
	KAIROS	0.991	0.818	1.000	0.900	0.995
	Ours	0.993	0.900	1.000	0.947	0.996
E3-CADETS	Threatrace	0.990	0.900	0.990	0.943	-
	FLASH	0.997	0.928	0.995	0.962	0.989
	KAIROS	0.994	0.800	1.000	0.889	0.997
	Ours	1.000	1.000	1.000	1.000	1.000
E3-ClearScope	KAIROS	0.958	0.000	0.000	0.000	0.500
	Ours	0.992	0.833	1.000	0.909	0.996
E5-THEIA	Threatrace	0.990	0.700	0.920	0.795	-
	KAIROS	0.983	0.714	1.000	0.833	0.991
	Ours	1.000	1.000	1.000	1.000	1.000
E5-CADETS-	Threatrace	0.970	0.630	0.860	0.727	-
	KAIROS	1.000	1.000	1.000	1.000	1.000
	Ours	1.000	1.000	1.000	1.000	1.000
E5-CADETS+	KAIROS	0.965	0.438	1.000	0.609	0.982
	Ours	0.996	1.000	0.857	0.923	0.929
E5-ClearScope	KAIROS	0.978	0.667	1.000	0.800	0.989
	Ours	0.993	0.833	1.000	0.909	0.997

scores even after behavior returns to normal. The CADETS-E5 results are reported under two views: E5-CADETS+ treats such post-attack residues as benign, while E5-CADETS- counts them as attacks and *FuzzyWatcher* sustains high recall with controlled false positives under both settings.

1) *Comparison Study of Detection*: We compare six representative PIDS: StreamSpot [34], Unicorn [2], Threatrace [10], MAGIC [8], KAIROS [3], FLASH [6]. StreamSpot, Unicorn and MAGIC are graph-level detectors, Threatrace and FLASH are node-level, and KAIROS supports graph- and batch-level operation. Each baseline is evaluated at its native granularity, and some cannot be run on all datasets due to label availability. *FuzzyWatcher* is run in both window- and graph-level modes. We report Accuracy, Precision, Recall, F1 and AUC, and additionally FPR and EER for graph-level experiments. For all baselines, we follow the configurations and thresholding strategies reported in the original papers without additional re-tuning on our datasets, to avoid introducing bias in cross-method comparison.

On DARPA, as reported in Table II, *FuzzyWatcher* is consistently competitive with or superior to prior node-level detectors. On E3-THEIA, FLASH attains the highest F1 of 0.956, while *FuzzyWatcher* achieves Accuracy 0.993 and the best AUC 0.996. On E3-CADETS, it reaches 1.000 on all metrics and clearly improves over Threatrace, FLASH, and KAIROS. On ClearScope, where attacks are sparse and subtle, KAIROS either misses all threats on E3 or suffers from low precision on E5, whereas *FuzzyWatcher* attains F1 around 0.90 on both E3 and E5 by leveraging uncertainty-aware representations and multi-criteria scoring. On the more challenging E5 datasets, it achieves 1.000 on all metrics for THEIA and CADETS-, surpasses Threatrace and matches KAIROS, and on CADETS+ it trades a slightly lower AUC for much higher precision and F1, 1.000 and 0.923 versus 0.438 and 0.609. At the graph level on StreamSpot, as reported in Table III, it matches KAIROS with perfect Accuracy, Precision, Recall, F1, and AUC and zero FPR and EER, while

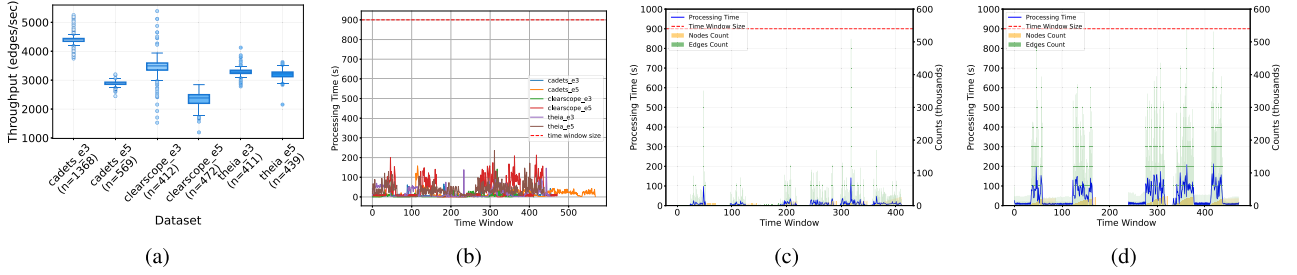


Fig. 3. Runtime evaluation. (a) Distribution of processing throughput (events/s) across evaluation datasets; (b) Per-window processing time (s). Red dashed line marks the time-window duration; (c–d) Per-window processing time and event counts for E3/E5-ClearScope, showing correlation between processing peaks and window size.

TABLE III

GRAPH-LEVEL ANOMALY DETECTION PERFORMANCE IN STREAMSPOT

Method	Acc	Prec	Rec	F1	AUC	FPR	EER
StreamSpot	0.870	0.843	0.910	0.875	0.897	0.170	0.125
Unicorn	0.935	0.886	0.997	0.939	0.925	0.125	0.108
MAGIC	0.986	0.983	0.989	0.986	0.977	0.017	0.008
KAIROS	1.000	1.000	1.000	1.000	1.000	0.000	0.000
Ours	1.000	1.000	1.000	1.000	1.000	0.000	0.000

TABLE IV

NODE-LEVEL ATTACK DETECTION PRECISION (ADP) IN DARPA DATASETS

	Threatrace	MAGIC	KAIROS	FLASH	Ours
E3-THEIA	0.03	0.00	0.45	0.02	0.28
E3-CADETS	0.01	0.03	0.01	0.10	0.16
E3-Clearscope	0.02	0.02	0.42	0.03	0.45
E5-THEIA	0.05	0.20	0.17	0.01	0.17
E5-CADETS	0.00	0.05	0.02	0.04	0.30
E5-Clearscope	0.01	0.01	0.01	0.33	0.39

TABLE V

SUMMARY OF EXECUTION TIME(S), AVERAGE THROUGHPUT(EVENTS/S) AND AVERAGE MEMORY OVERHEAD(MB)

Dataset	Med Time	Avg Thr. ev/s	Avg Mem. MB
E3-CADETS	6.8	4398.4	213.99
E5-CADETS	9.5	2887.7	214.42
E3-ClearScope	1.9	3465.7	214.55
E5-ClearScope	12.2	2369.9	214.66
E3-THEIA	8.7	3284.1	213.48
E5-THEIA	27.7	3201.8	213.73

outperforming statistical and static baselines, which indicates that its dynamic and context aware reasoning scales from local windows to whole-graph anomalies.

Across Section V-B, anomaly magnitude alone is insufficient in noisy conditions. Perfect recall can coexist with reduced F1 due to residual false positives, as on E5-ClearScope. Incorporating rarity aware and propagation aware cues reduces these residues while preserving recall and AUC, thereby improving precision and overall dependability for operational use.

2) *Node-Level Attack Detection Precision*: The compared PIDSs operate at different detection granularities, a single operating threshold may give a biased view of their effectiveness, so a unified threshold is not enforced. To enable a fair and threshold-independent comparison at a common node-level view, we follow [37] and adopt the Attack Detection Precision (ADP) metric together with unified node labels. Unlike threshold-dependent metrics, ADP evaluates detection performance over the entire ranking of anomaly scores rather than at a single operating point, making the comparison invariant to threshold selection and less sensitive to differences in score distributions across methods. ADP quantifies, over all attainable precision levels, how rapidly a detector surfaces distinct attacks when nodes are ranked by anomaly score. We project all methods to this unified node-level setting and compare their ADP scores under the same alert ranking scenario. The results are summarized in Table IV.

Across the DARPA, the ADP comparison shows that our method surfaces attacks earlier, or at least competitively, under a unified node ranking. On E3-CADETS, E3-ClearScope, E5-CADETS and E5-ClearScope, our detector achieves the highest ADP. KAIROS obtains the best score on E3-THEIA and remains competitive on E5-THEIA, while MAGIC attains the highest score on E5-THEIA but lags behind on the other datasets. A plausible explanation is that THEIA contains more spatially and temporally scattered attack nodes with fewer long, coherent malicious paths, which reduces the relative advantage of path-oriented reasoning. Threatrace and FLASH generally show lower ADP, suggesting that their alerts are either concentrated on a small subset of attacks or more heavily mixed with benign nodes at the top of the ranking. These results indicate that, under a common node-level view and without per-method threshold tuning, *FuzzyWatcher* delivers more consistent and efficient attack discovery across heterogeneous DARPA scenarios.

C. Runtime and Deployment

We evaluate runtime behavior through the time window. Fig. 3(a) summarizes per window throughput across all evaluation traces and Fig. 3(b) plots per window processing times, with a red dashed line marking the 15,min window boundary. Table V reports processing times and average throughput in events per second for each dataset.

During the detection phase *FuzzyWatcher* processes almost all windows efficiently. Across various traces the median per window processing time is approximately 9 s while the median throughput reaches 3.4 k events/s as shown in Table V. Main memory usage remains minimal at an average of 214 MB with no GPU across all datasets. Although processing costs increase with window size the latency stays stable for most windows. Rare spikes in processing time shown in Fig. 3(c) and (d) coincide

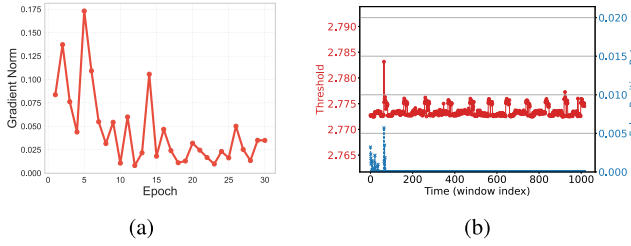


Fig. 4. Stability Analysis. (a) Gradient stability. (b) Long-term deployment stability.

with high graph density and dominate the upper tail. Nevertheless almost every window finishes well within the 15 min budget which provides sufficient slack for real time processing even during occasional workload surges. This runtime profile is consistent with practical deployment settings.

D. Stability

We empirically validate the boundedness and gradient stability of DFMA and examine the behavior of the POT threshold in a streaming deployment.

Fig. 4(a) reports the ℓ_2 norm of the DFMA parameters' gradients over 30 training epochs. The norm is larger and more variable in the first few epochs, then quickly decreases and stays at a small magnitude without spikes. This monotone decay and the absence of large fluctuations indicate a stable optimisation process without gradient explosion or vanishing, and show that DFMA converges as expected under the boundedness guarantees discussed in Section IV-B2. Fig. 4(b) evaluates threshold reuse and detection stability in a long streaming run. We process a continuous sequence of 1,016 windows with 20,480 edges each, spanning more than twenty million events. At the start we reset the model memory to simulate a cold start, then handle all windows in time order with a fixed POT configuration. The threshold remains in a narrow band around a mean of about 2.774, with POT adapting slightly to score changes while staying overall stable. A few spikes appear in the first 60 to 100 windows during warm up after the memory reset, after which the false positive rate is close to zero for more than 900 windows. This indicates that after a cold start *FuzzyWatcher* quickly stabilises and, with automatically updated yet stable thresholds, maintains a low false positive rate over long continuous runs at the time scale of the evaluated traces. These results show that the method can operate continuously under evolving data streams without threshold re-tuning, while controlling false positives after initial warm-up, which is consistent with the requirements of real audit systems.

We also evaluate robustness to adversarial perturbations on node features and graph structure. As shown in Fig. 5, the test AUC remains consistently near 0.99 while feature noise and edge flip rates increase from 0.0 to 0.3. The performance decreases only slightly from 0.9904 in the clean model to 0.9804 in the worst case scenario. Although feature noise exhibits a marginally stronger impact than edge flipping, the combined effect of these perturbations results in a minimal absolute drop. These results show that *FuzzyWatcher* is stable under moderate

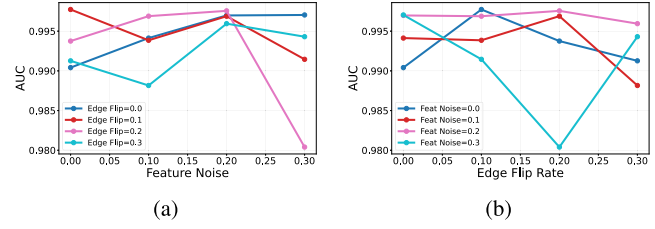


Fig. 5. Adversarial robustness test. (a) AUC under different feature noise. (b) AUC under different edge flipping rates.

corruption of both attributes and connectivity, which supports deployment in noisy and potentially adversarial environments.

E. Parameter Sensitivity Analysis

Fig. 6 studies the effect of key hyperparameters. Fig. 6(a) varies the number of DFMA rules on E3–ClearScope. Training time grows roughly linearly with the number of rules, while prototype correlations drop sharply when moving from two to eight rules and then improve more slowly. Projection correlations are already very small in all settings. A simple composite score that balances training time and prototype and projection independence selects eight rules as a good trade off between efficiency and diversity. Fig. 6(b) scans the reconstruction weight λ and the dropout rate on E5–THEIA. The model is stable over a wide range, with validation AUC above 0.998 and test AUC between 0.986 and 0.995 for λ from 0.0 to 0.6, whereas setting λ to one, that is using only reconstruction loss, causes performance to collapse. Varying dropout values around 0.2 to 0.3 give better F1. Fig. 6(c) varies the historical neighbour size and the POT hyperparameters μ and q . A shallow temporal depth with one historical step and up to sixteen neighbours already reaches the best AUC. There is a broad safe region with μ between 90 and 95 and q between 0.85 and 0.95 where AP remain near their optimum (see the supplementary analysis for other metrics). Only very aggressive settings such as $q = 0.98$ hurt recall. *FuzzyWatcher* achieves high performance without fine tuning and shows robust behaviour across a wide range of hyperparameter choices.

F. Ablation Study

As Fig. 7(a) shows, the ablation study quantifies the contribution of the two uncertainty modules. The complete model attains Precision 88.4%, Recall 100%, F1 93.6% and AUC 99.8%; removing FCEM cuts Precision to 56.5% (F1 63.1%), while removing DFMA lowers Recall and AUC (Recall 76.1%, AUC 87.4%). Dropping both produces the weakest outcome (Precision 49.7%, Recall 78.6%, F1 56.8%, AUC 87.9%), showing that the two components are complementary: FCEM primarily suppresses false alarms, DFMA recovers weak but persistent signals. Our benign-only, self-supervised training deliberately favors recall to avoid missed campaigns. However, this recall bias can raise false positives and therefore increase analysts' workload, an effect that FCEM helps mitigate.

Fig. 7 reports training-loss curves for the ablation study. The full model not only achieves the best detection metrics but also

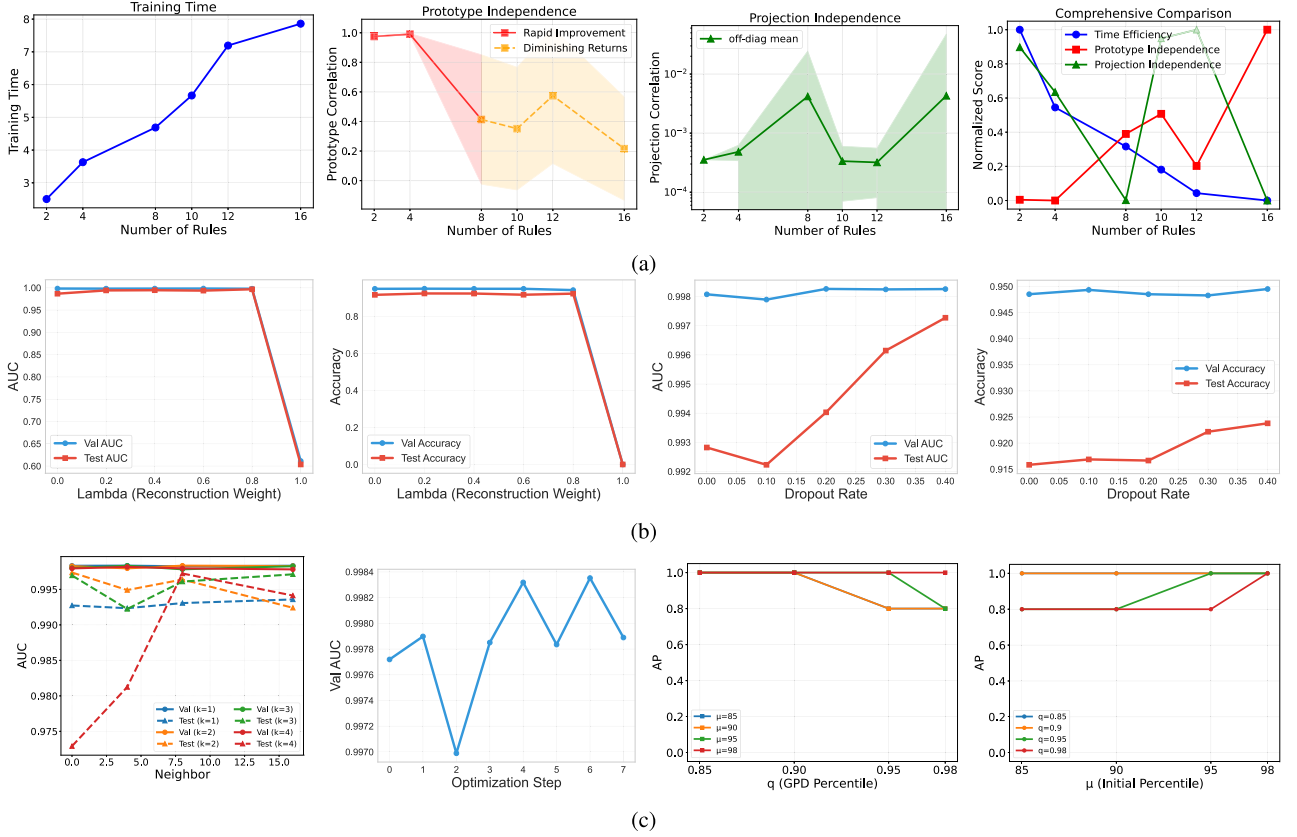


Fig. 6. Model parameter analysis (a) The number of rules (E3-Clearscope). (b) Reconstruction loss weight and Dropout rate (E5-THEIA). (c) Historical Neighbor Size (E5-THEIA) and threshold-adaptive hyperparameters (E3-CADETS).

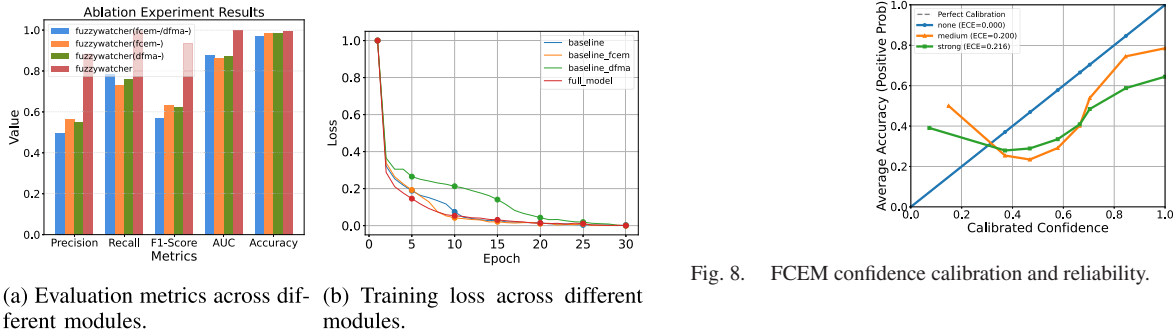


Fig. 7. Ablation study.

converges stably to the lowest training loss. Removing DFMA slows convergence and increases volatility, whereas removing FCEM harms early-stage stability and false-positive control. These curves indicate that the proposed modules do more than adjust scoring at inference: they improve representation learning and training stability. Together with the reduction in inference time candidates, these properties point to practical efficiency for large-scale deployments.

VI. DISCUSSION

Empirically, *FuzzyWatcher* improves aggregate metrics and triage-time dependability. At time-window granularity with a calibrated threshold, it consistently shrinks the candidate set

Fig. 8. FCEM confidence calibration and reliability.

from tens of thousands of windows to a few hundred while maintaining high recall, materially reducing analyst workload without sacrificing coverage. Benign-only training further suits label-limited settings. These gains arise from (i) *uncertainty-aware* FCEM weighting that de-emphasizes ambiguous or rare-but-benign patterns, and (ii) *fuzzy-guided temporal aggregation* DFMA that regularizes temporal context. Together they widen benign/abnormal separation while preserving sensitivity to low-frequency attack traces.

To ensure operational utility, FCEM scores should function as interpretable epistemic uncertainty estimates rather than opaque weights. We assess whether FCEM behaves as a calibrated confidence signal using Expected Calibration Error (ECE) and the Brier score under three settings (Fig. 8, detailed in Supp. C). On in-distribution validation data, low ECE and Brier scores, together with a near-diagonal reliability curve, indicate that

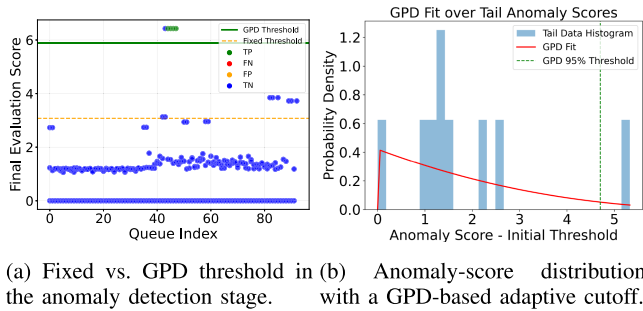


Fig. 9. Adaptive threshold on E3-CADETS: fixed vs GPD.

FCEM values align with uncertainty. Under medium and strong drift, both ECE and Brier increase only moderately, while the reliability curves remain smooth and monotone. High FCEM scores correspond to rare or out-of-distribution cases, whereas low scores align with well-supported patterns. These results show that FCEM provides stable uncertainty estimates and preserves the relative ordering of risk under schema shifts. This means that the model's decision signals remain consistent when data distributions change, without requiring recalibration. As a result, FCEM continues to provide consistent uncertainty estimates across distribution changes, maintaining comparable scores for similar levels of support. This supports stable behavior under evolving system conditions.

A practical detector should expose a tunable operating point with interpretable calibration. Fixed thresholds are brittle under workload drift, whereas GPD on tail risk provides a data-adaptive decision boundary. In our results Fig. 9, GPD-based thresholds suppress spurious spikes and align decisions with propagation-aware cues from temporal neighborhoods, thereby correcting borderline cases. This suggests that *calibrated score distributions combined with propagation semantics* are a robust recipe for deployment.

Real deployments face cross host or domain shifts and non IID observations. Our framework alleviates this by using FCEM to bias representation learning toward rare but benign structures observed during training while aggregating multi scale structure so decisions do not hinge on transient cues. While these results are promising, we acknowledge that a full task oriented user study in a production Security Operations Center remains beyond the scope of this paper. Measuring specific metrics such as the time to first correct triage or the exact reduction in false positive effort requires complex longitudinal logistics. Therefore, evaluating the integration of our auditable signals with realistic analyst workflows and conducting controlled user studies represents an important direction for future work to further validate the deployability of *FuzzyWatcher* across heterogeneous estates.

VII. CONCLUSION

This work proposes *FuzzyWatcher*, a graph-learning framework for provenance-driven continuous anomalous behavior detection in modern attacks. The method couples a memory-based temporal encoder for continuous-time interactions with fuzzy neuron that adaptively modulates representations under uncertainty. By integrating historical and structural context with

multi-criteria anomaly scoring, the framework targets dependable detection in non-stationary, anomaly label-limited environments. Experiments show consistent performance and stable generalization under benign-only training, while the interpretable, the rules of fuzzy neuron partitioning supports reliable decisions during investigation. The design emphasizes operational practicality: window-level modeling narrows the search space for analysts, and uncertainty-aware aggregation maintains sensitivity to rare yet benign patterns without reliance on hand-crafted rules. However, our current design focuses exclusively on epistemic uncertainty and does not explicitly model aleatoric noise in audit logs, which remains an important limitation. In future work, we will investigate cross-environment adaptation across different audit schemas, entity types, and system semantics to improve transferability to heterogeneous real-world deployments.

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